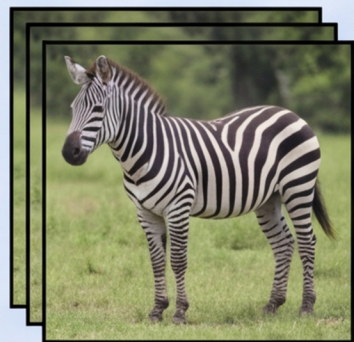
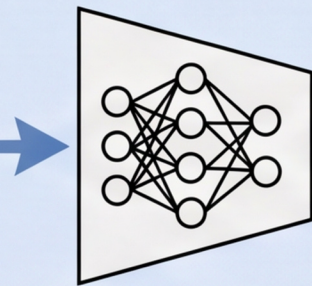


Phase 1: Neural-to-semantic transition

Input representations
and latent matrix

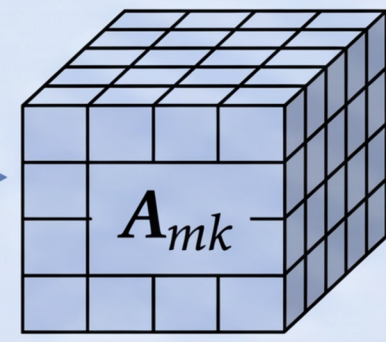


Training image
(AwA2) benchmark



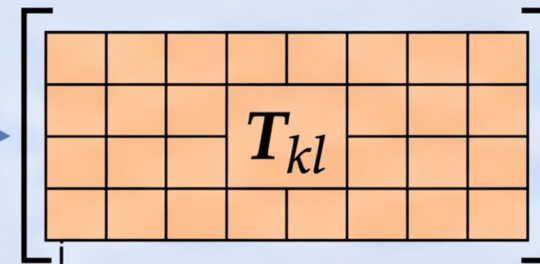
Frozen ResNet-101
feature extractor

Uninterpretable
latent feature space



m = training dataset universe objects
 k = 2048-dimensional neural feature

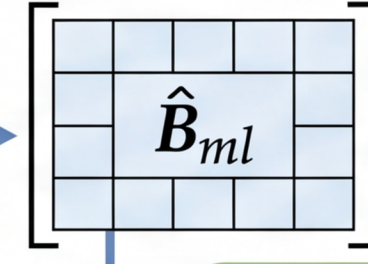
Global transition
operator



Rank- r SVD basis V_r

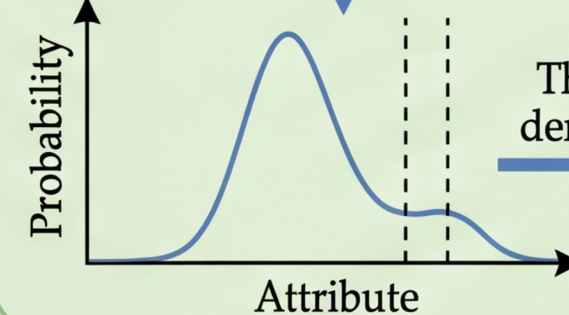
Auxiliary principal component PA
compresses collinear neural features
apply linear ridge regression weights

Reconstructed continuous
semantic attributes



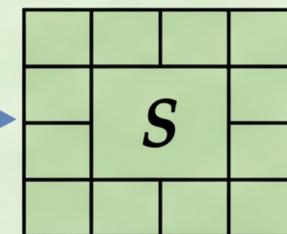
Phase 2.1: Semantic attribute selection and stable discretization (WEDD)

Physical conceptual salience



Threshold
density cuts

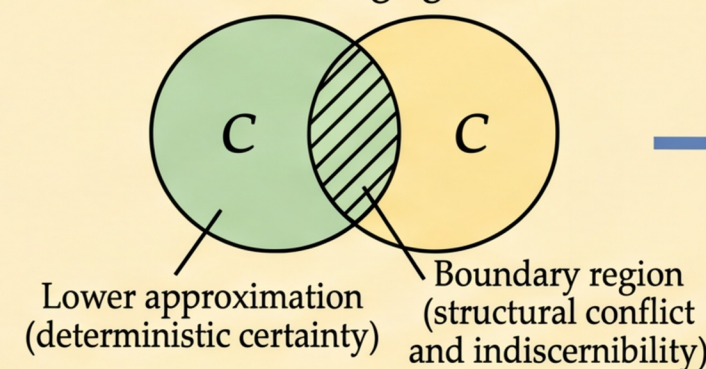
Discrete symbolic
states



$$\hat{B} = (A - \bar{1})^T T + \bar{1} \cdot b_0^T$$

Phase 2.2: Rough-set granulation and knowledge discovery

Formal knowledge granulation



Global conflict-aware production
rulebook $\mathcal{R}$

$$r : \bigwedge_{j \in R_r} [s_j = v_j] \Rightarrow d = y_r$$

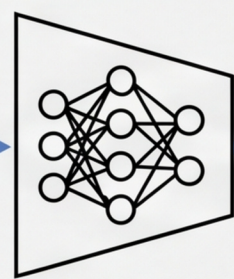
IF paws = s2 AND hunter = s2 AND small = s0
AND yellow = s2 $\Rightarrow$
class: tiger (confidence 0.896)

Phase 2.3: Conflict-aware inference and evaluation

Static rules induced audit real-time unobserved data



Test image



Frozen ResNet-101

$a \times k$
unseen test
feature vector

Project via
transition weights
 T

$\hat{b} \times l$
unseen
continuous
semantic

Apply WEDD
density cuts

s
active test
symbolic
signature

Query gate
Global
rulebook $\mathcal{R}$



Exact match: Covered fidelity and covered
accuracy (verifiable logical subset)



Multiple rule firing: Resolve via
integral support score $S(y_k)$



Structural gap: Soft match via Hamming
distance d_H or trigger audit abstention